

General information about the exercises

Data set:

In the exercises for this Reading Course, we use the `Wage` data set which is available in the R package `ISLR`. It contains information about 3000 men from the Mid-Atlantic region of the United States.

You can find more information about this data set here:

<https://rdrr.io/cran/ISLR/man/Wage.html>

and in the book

James, G., Witten, D., Hastie, T., and Tibshirani, R. (2013) An Introduction to Statistical Learning with applications in R, <https://www.statlearning.com>, Springer-Verlag, New York.

For our analysis of the data set, we discard the columns `region` and `logwage`.

Solutions:

There will be no sample solutions. If you worked on the exercises and are not sure whether your solutions are correct, you can e-mail me your solution and I will give you feedback. Note that you will only get feedback if you provide your attempt first!

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Exercise 1 (Simple Linear Regression):

We are interested in the effect of an individual's age on his wage.

- (a) Specify the model equation.
- (b) According to the method of least squares, the parameter estimates for the model are $\hat{\beta}_0 = 81.70$ and $\hat{\beta}_1 = 0.71$. Interpret the estimated slope parameter.
- (c) Why should $\hat{\beta}_0$ not be interpreted here?

Exercise 2 (Multiple Linear Regression):

We now also include the education level to our model from Exercise 1.

- (a) Define the dummy variables for the education levels and specify the model equation.
- (b) Consider the following R output for the model from (a):

```
Coefficients:
              Estimate
(Intercept)  60.33597
age           0.56869
education2. HS Grad  11.43865
education3. Some College  24.16700
education4. College Grad  39.76677
education5. Advanced Degree 64.98656
```

Interpret the effect(s) of education on the wage. Which category was used as reference category?

- (c) What is the estimated wage difference between a 50 year old man with education level 3 (Some College) and a 50 year old man with education level 5 (Advanced Degree)?
- (d) What is the estimated wage difference between a 40 year old man with education level 4 (College Grad) and a 20 year old man with education level 1 (< HS Grad)?